STRATEGIC ACADEMIC RESTRUCTURING & CURRICULUM REFORM

Faculty of Geoscience and Technology

An integrated model for Earth science, mineral and energy exploration, civil engineering, and space and geospatial technology.

STRATEGIC ADVISORY REPORT • NOVEMBER 2026

Prepared to support review by the Dean and Faculty Academic Council.

EXECUTIVE SUMMARY

An integrated faculty with clear academic responsibilities

The Faculty of Geoscience and Technology has a distinctive opportunity: to combine rigorous Earth science with applied engineering, resource development and space-enabled observation. Its present research portfolio is broad, but the relationships among disciplines, laboratories and degree programs are not always explicit. This proposal keeps the integrated identity while clarifying educational ownership and reducing curriculum overlap.

Central recommendation

Create **five academic departments** supported by **six shared laboratories** and a **faculty-wide Drilling and Subsurface Engineering Center**. Organize undergraduate education through common scientific foundations, discipline-specific cores, professional specializations and integrated final-year projects.

- **Organization:** Earth Sciences and Mineral Exploration; Applied Geophysics and Geochemistry; Mineral, Mining, Petroleum and Drilling Engineering; Civil and Geological Engineering; Space Science and Geospatial Technology.
- **Majors:** Twelve detailed academic areas across geoscience (5), engineering (4), and space/geospatial technology (3). Approximately eight to nine should be considered for initial degree status; remaining areas may be tracks.
- **Curriculum:** Separate compulsory common foundations from major-specific subjects; teach shared content once; strengthen fieldwork, laboratories, computing and professional design.
- **Integration:** Introduce twelve interdisciplinary subjects, including integrated mineral exploration, subsurface modeling, satellite remote sensing and multidisciplinary capstone design.
- **Execution:** Use a phased three-to-five-year transition with curriculum auditing, staff and laboratory planning, industry engagement and quality assurance.

Terminology. *The original term "universe" is interpreted here as space science and space engineering. "Exporters" is interpreted as experts or highly qualified specialists. All organizational recommendations require validation against national regulations, staffing, facilities and finances.*

REPORT GUIDE

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How to use this report

Sections 1-3 explain the reform case and proposed organization; Sections 4-7 provide degree architecture and a detailed course catalogue; Sections 8-10 translate the proposal into multidisciplinary education, implementation and governance. The final section lists the official university references consulted.

1. Background and academic development objectives

The university offers four-year undergraduate education in science and engineering. The Faculty presently addresses geology and ore deposits; mineral, coal, petroleum and gas exploration and extraction; geophysics and geochemistry; drilling; civil investigation and construction; groundwater; surveying; space-related disciplines; tourism-area development; and industrial mineral and rock applications.

Its current research facilities are six laboratories—Resource Exploration, Civil Engineering, Mineral Engineering, Geophysical Prospecting, Petroleum and Space—with a separate Drill Engineering Division. The proposed reform should preserve specialized research expertise while establishing a better distinction among academic disciplines, laboratories, degrees and professional tracks.

1.1 Main issues identified

- The faculty does not communicate a clear relationship between theoretical geoscience and professional engineering.
- Existing laboratories represent a mixture of disciplines, application areas and equipment-oriented specializations.
- All majors receive a common geology-heavy core even when physics, mechanics, chemistry or computing would be more relevant.
- Growth of electives has increased duplication across majors, constraining practical and integrative teaching time.

1.2 Educational mission

To cultivate internationally competitive geoscientists and engineers with strong scientific foundations, advanced technological knowledge, practical professional capabilities, and interdisciplinary problem-solving skills.

Three complementary missions are recommended: **understand the Earth**; **develop and utilize Earth resources and infrastructure**; and **understand space and use geospatial technology**. These missions should be connected by shared mathematics, physical sciences, computing, data methods, ethics and engineering practice.

2. International university benchmarking

Leading universities use more than one institutional model. Some separate Earth sciences from civil and aerospace engineering; others bring related sciences together while retaining distinctive professional degrees. The strongest general lesson is to preserve disciplinary depth while making cross-disciplinary teaching and research easy.

Institution	Organizational / educational pattern	Lesson for the faculty
Massachusetts Institute of Technology	Separate EAPS and Civil and Environmental Engineering departments; joint climate systems science and engineering degree illustrates cross-departmental teaching.	Distinct departments can co-own interdisciplinary programs. [R1]
University of Cambridge	Earth Sciences operates in the Natural Sciences Tripos, with quantitative study and substantial fieldwork.	Strong foundations and structured field training. [R2]
University of Oxford	Earth Sciences combines geological theory, quantitative methods and fieldwork.	Progression from fundamentals to advanced interpretation and independent work. [R3]
ETH Zurich	Earth and Planetary Sciences and Civil, Environmental and Geomatic Engineering are separate departments; interdisciplinary centers connect fields.	Clear departmental responsibilities with shared interdisciplinary opportunities. [R4]
Peking University	School of Earth and Space Sciences includes geology, geochemistry, geophysics, space science and geographic information; four-year programs share first-year foundations.	Integrated Earth-and-space science, followed by specialization. [R5]

Institution	Organizational / educational pattern	Lesson for the faculty
Jilin University	Geoexploration programs include geophysics, exploration technology and engineering, GIS and surveying/mapping.	Explicit separation of exploration techniques and measurement technologies. [R6]
University of Tokyo	Earth and Planetary Sciences, civil engineering and aerospace engineering are housed in distinct academic units.	Maintain disciplinary identities across science and engineering.
Tsinghua University	Civil and aerospace disciplines follow specialized engineering pathways distinct from Earth system research.	Strong engineering professional cores.
University of California, Berkeley	Civil and environmental engineering includes geotechnical and geo-systems activities.	Geoscience expertise can support engineering education.
Colorado School of Mines	School of Earth, Space and Planets and School of Minerals and Subsurface Energy connect geological/geophysical engineering with mining and petroleum.	Highly relevant integrated resource-and-engineering example. [R7]

2.1 Implications for the faculty

- Do not insist that every discipline belong to one broad geology department.
- Preserve a common institutional mission while distinguishing scientific departments from engineering departments.
- Give each program its own required professional foundation; use shared facilities and cross-listed modules for common techniques.
- Integrate education through problem-based work, not repeated introductory content.

3. Assessment of the current system

3.1 Strengths worth preserving

- The capacity to follow a resource project from geological understanding through exploration, drilling, mining and processing.
- Natural links between geological site investigation, geotechnical engineering and civil design.
- Opportunity to apply satellite, surveying, GIS and remote-sensing methods across mineral exploration and infrastructure.
- Existing laboratory capabilities can be shared among degrees without duplicating major capital investments.

3.2 Academic weaknesses requiring action

- **Unclear boundaries:** mineralogy/petrology is fundamental science, whereas mineral materials and rock applications are engineering/materials fields.
- **Methods mistaken for organizational divisions:** electromagnetic and magnetic prospecting may be specialized research groups within a broader geophysics discipline.
- **Insufficient visibility:** resource estimation, hydrogeology, geotechnical engineering, mine planning, processing, reservoir engineering and satellite data science need a stronger academic position.
- **Curricular duplication:** identical foundational geology courses should not be compulsory at the same depth for civil, geophysics, petroleum and space students.

Organizing principle

Organize departments by **disciplinary responsibility**, degree programs by **graduate competencies**, and research projects by **real-world problems**. Laboratories are enabling facilities rather than substitutes for academic departments.

4. Recommended faculty organization

The proposed faculty has five departments, linked through faculty-wide committees and shared research and education facilities.

1. Earth Sciences and Mineral Exploration

General and Structural Geology; Mineralogy and Petrology; Economic Geology and Ore Deposits; Mineral Exploration and Resource Evaluation; Petroleum Geology and Basin Analysis; Hydrogeology and Groundwater Science.

2. Applied Geophysics and Geochemistry

Seismic Exploration; Gravity and Magnetic Exploration; Electrical and Electromagnetic Exploration; Exploration and Environmental Geochemistry; Instrumentation; Data Processing, Inversion and Interpretation.

3. Mineral, Mining, Petroleum and Drilling Engineering

Mining Engineering; Mineral Processing; Extractive Metallurgy and Mineral Materials; Petroleum Engineering; Drilling and Subsurface Engineering; Mining and Petroleum Machinery and Automation.

4. Civil and Geological Engineering

Structural and Building Engineering; Geotechnical and Foundation Engineering; Engineering Geology and Site Investigation; Underground and Tunnel Engineering; Transportation and Infrastructure; Water Resources; Construction Management.

5. Space Science and Geospatial Technology

Space Physics and Space Environment; Atmospheric Science and Earth Observation; Planetary Science; Satellite and Space Systems Engineering; Surveying/Geodesy/GNSS; Remote Sensing and GIS; Geospatial Data Science and Spatial Planning.

4.1 Teaching roles versus laboratory roles

Departments should own curriculum, academic standards, instructional staffing, program review, degrees and professional outcomes. **Laboratories** should deliver experiments, applied research, field instruments, computing facilities, analytical services and shared student training.

Facility	Revised shared role
Resource Exploration	Geological mapping, deposits, resource evaluation, geochemical analysis and common field geology facilities.

Facility	Revised shared role
Civil Engineering	Structural materials, geotechnical investigation, foundations and infrastructure design/testing.
Mineral Engineering	Mineral processing, extractive metallurgy, mineral materials and rock utilization.
Geophysical Prospecting	Seismic, gravity, magnetic, electrical/EM methods, instruments and data processing.
Petroleum	Drilling, reservoir/production engineering, petroleum machinery and processing facilities.
Space	Space science, atmospheric science, satellite data, Earth observation and geospatial information.
Drilling & Subsurface Engineering Center	Faculty-wide hub for drilling equipment, borehole measurements, drilling fluids and subsurface site investigations.

4.2 Transfer or cross-appointment of existing divisions

- Mineralogy and Petrology: academic ownership in Department 1, laboratory access shared with mineral engineering.
- Geochemical Prospecting: academic ownership in Department 2.
- Petroleum Prospecting/Geology: Department 1; petroleum production, machinery and processing: Department 3.
- Independent Drill Engineering Division: academic home in Department 3 with a faculty-wide shared center.
- Surveying, GIS and satellite imaging: Department 5, cross-taught in Civil and Geological Engineering.
- Tourist District Development: interdisciplinary geotourism/geospatial planning specialization; not initially a stand-alone engineering degree.

4.3 Faculty-wide governance

Academic and Curriculum Committee. Set learning outcomes, approve course ownership, remove duplication, monitor prerequisites and oversee annual program review.

Interdisciplinary Research and Education Committee. Coordinate cross-departmental projects in mineral exploration, resource engineering, infrastructure, planetary studies and satellite monitoring.

Industry Advisory Committee. Include geological surveys, mining/petroleum employers, construction firms, geospatial/space providers and relevant professional bodies for curriculum and internship feedback.

5. Proposed degree programs and detailed major areas

The proposal defines **12 detailed areas** grouped into three academic sectors. The 12 areas should not automatically become 12 independent undergraduate degrees: offer approximately eight to nine principal degrees initially, depending on institutional permissions, staffing and laboratory capacity.

5.1 Geoscience and exploration

Code	Detailed major area	Main content
G1	Geology and Mineral Exploration	Geological mapping, structural geology, economic geology, ore deposits, exploration and resource estimation.
G2	Applied Geophysics	Seismic, gravity, magnetic, electrical/EM, borehole geophysics and integrated interpretation.
G3	Applied Geochemistry	Exploration geochemistry, analytical and environmental geochemistry and geochemical data analysis.
G4	Petroleum Geoscience	Basin analysis, petroleum systems, reservoir geology, sedimentology and exploration.
G5	Hydrogeology and Environmental Geoscience	Groundwater occurrence/flow/quality, subsurface hazards and environmental geology.

5.2 Engineering and technology

Code	Detailed major area	Main content
E1	Geological and Drilling Engineering	Site investigation, rock mechanics, geological engineering, drilling technology and borehole engineering.
E2	Mining and Mineral Processing Engineering	Mine design, extraction, mineral separation, metallurgy, safety and mineral materials.
E3	Petroleum Engineering	Well design, drilling/completion, reservoirs, production, petroleum facilities and management.
E4	Civil and Geotechnical Engineering	Structural design, foundations, tunnels, transportation, water and construction management.

5.3 Space science and geospatial technology

Code	Detailed major area	Main content
S1	Space Science and Technology	Space physics, planetary science, atmosphere, space environment and exploration.
S2	Space Systems and Satellite Engineering	Orbital mechanics, spacecraft design, satellite subsystems and mission engineering.
S3	Geomatics and Geospatial Engineering	Surveying, geodesy/GNSS, GIS, photogrammetry, remote sensing and spatial analysis.

5.4 Suggested initial degree packaging

A practical first phase is to treat **G1, G2, G3, E1, E2, E3, E4, S1 and S3** as nine possible main undergraduate degrees. G4 and G5 can initially be specialization tracks within G1, and S2 a track within S1. This is an academic planning option, not a determination of legal or accreditation eligibility.

Geotourism, Geoheritage and Regional Development should be a cross-disciplinary specialization involving geospatial engineering, geology and civil infrastructure, including geoparks, spatial planning and environmental assessment.

6. Four-level curriculum framework

Replacing a geology-heavy common core with an outcomes-based four-level curriculum is the principal efficiency improvement. All students need scientific literacy, but only particular majors require advanced mineralogy, mechanics, electrical engineering or orbital dynamics.

Level I — Common science and engineering foundation. Calculus, physics, chemistry, programming, statistics, communication and appropriate introductions to mechanics, materials, electronics or computing.

Level II — Discipline-specific foundations. Geological mapping and petrology for geoscience; physics and signal processing for geophysics; chemistry and analytical methods for geochemistry; mechanics for engineering; geodesy and electromagnetism for space/geospatial majors.

Level III — Professional specialization. Exploration, resource estimation, mine design, drilling, petroleum reservoirs, civil structures, satellite systems and GIS according to degree requirements.

Level IV — Integration and professional practice. Projects, field camps, laboratory/design work, cross-disciplinary electives, internships, professional ethics and a capstone thesis/design project.

6.1 Indicative credit distribution



This allocation is illustrative: 10% general education, 20% mathematics/science/computing, 20% disciplinary foundations, 30% major-specific subjects, 20% applied practice/internship/capstone. It must be reconciled with national degree rules and the university credit system.

6.2 Different majors require different compulsory foundations

Major	Required foundation emphasis
Geology / mineral exploration	Physical geology, mineralogy, petrology, structural geology, stratigraphy and field mapping.
Applied geophysics	Mathematics, physics, geology, geophysical theory, data processing and field measurements.
Applied geochemistry	Chemistry, mineralogy, petrology, geochemistry, analytical chemistry and sampling.
Geological / drilling engineering	Engineering geology, mechanics, fluids, rock mechanics and drilling fundamentals.
Mining / mineral processing	Mineralogy, rock mechanics, mine methods, mineral processing and materials science.
Petroleum engineering	Petroleum geology, fluids, thermodynamics, rock properties and engineering mechanics.
Civil and geotechnical	Engineering mechanics, mechanics of materials, soils, structural concepts and engineering geology.
Space science and technology	Advanced mathematics, physics, electromagnetism, computing and atmospheric/space physics.
Geomatics / geospatial	Mathematics, statistics, programming, surveying, geodesy and GIS.

7. Recommended subject catalogue and descriptions

The following courses represent a **proposed subject bank**. They are not all compulsory for any one degree. Each department should define required courses, prerequisites, credit value, contact hours, practical work and assessment for the relevant majors.

7.1 Common scientific and engineering foundation

#	Proposed subject	Course summary
1	Calculus and Differential Equations	Differential and integral calculus, multivariable calculus, vector calculus and differential equations for science and engineering.
2	Linear Algebra and Numerical Methods	Matrix methods, eigenvalue problems, numerical algorithms, computational solutions and optimization.
3	Physics for Earth and Engineering	Mechanics, waves, electricity, magnetism, thermodynamics and Earth/engineering applications.
4	General and Analytical Chemistry	Chemical reactions, equilibrium, solution chemistry, analytical techniques and natural materials.
5	Programming and Scientific Computing	Programming, scientific algorithms, data analysis, numerical computation and visualization.
6	Probability, Statistics and Data Analytics	Statistical inference, regression, uncertainty analysis, experimental design and interpretation.
7	Engineering Mechanics	Statics, dynamics, force systems, equilibrium and mechanical behavior.
8	Fluid Mechanics and Thermodynamics	Fluid properties, flow, energy conservation, heat transfer and subsurface/hydraulic applications.
9	Electrical and Electronic Engineering	Electric circuits, electronics, sensors, instruments and measurement principles.
10	Materials Science and Engineering	Material structure, mechanical properties, tests, selection and engineering uses.
11	Engineering Drawing and Computer-Aided Design	Technical graphics, CAD, three-dimensional models and engineering design documentation.
12	Technical Communication and Professional Practice	Technical English, scientific writing, reports, engineering ethics, safety and teamwork.

7.2 Geoscience and exploration

#	Proposed subject	Course summary
1	Introduction to Earth System Science	Earth structure, composition and evolution; linked geosphere, atmosphere, hydrosphere and biosphere.
2	Physical Geology	Geological materials, processes, structures, surface processes and Earth evolution.
3	Crystallography and Mineralogy	Crystal structure, mineral composition and classification, optical and physical mineral identification.
4	Igneous and Metamorphic Petrology	Formation, composition, classification and evolution of igneous and metamorphic rocks.
5	Sedimentology and Stratigraphy	Sedimentary processes, depositional environments, stratigraphic relations and geological history.
6	Structural Geology and Tectonics	Deformation, faults, folds, fractures, tectonics and controls on ores and subsurface structures.
7	Geological Mapping and Field Methods	Field observation, measurement, sampling, mapping, cross sections and geological interpretation.
8	Economic Geology and Ore Deposits	Genesis, classification and geological controls of metallic and non-metallic mineral deposits.
9	Mineral Exploration and Resource Evaluation	Prospect planning, sampling, exploration methods, resource estimation and geological evaluation.
10	Applied Geophysics	Seismic, gravity, magnetic, electrical and electromagnetic methods for subsurface investigation.
11	Exploration Geochemistry	Geochemical sampling, analyses, anomalies, element mobility and mineralization interpretation.
12	Hydrogeology and Groundwater Science	Aquifers, groundwater flow, exploration, groundwater quality and resource assessment.
13	Petroleum Geology and Basin Analysis	Hydrocarbon generation/migration/trapping, sedimentary basins, reservoirs, seals and petroleum systems.
14	Geostatistics and 3D Geological Modeling	Spatial statistics, interpolation, uncertainty, geological grids and three-dimensional subsurface models.

7.3 Geological, drilling, mining and mineral engineering

#	Proposed subject	Course summary
1	Engineering Geology and Site Investigation	Geological characterization of construction sites, slopes, foundations and underground works.
2	Soil Mechanics and Foundation Engineering	Soil behavior, stress, consolidation, shear strength, bearing capacity and foundation concepts.
3	Rock Mechanics and Underground Engineering	Rock mass behavior, underground excavation stability, rock deformation and ground support.
4	Drilling Engineering and Well Design	Drilling methods, borehole planning, drilling performance, operations and well construction.
5	Drilling Equipment and Machinery	Drilling rigs, drill bits, pumps, equipment selection, maintenance and technology improvement.
6	Drilling Fluids and Well Control	Drilling fluid properties, circulation, borehole stability, pressure control and well safety.
7	Mine Design and Planning	Surface and underground mine layout, production schedules, equipment and optimization.
8	Mining Engineering and Operations	Mining methods, mine development, ventilation, machinery, safety and production.
9	Mineral Processing and Separation Engineering	Crushing, grinding, classification, gravity/magnetic separation, flotation and dewatering.
10	Extractive Metallurgy and Mineral Materials	Metals recovery, pyro-/hydrometallurgy, recycling and value-added mineral materials.
11	Mining Environmental Engineering	Waste and tailings, mine water, emissions, rehabilitation, closure and environmental protection.
12	Mineral Resource Economics and Project Evaluation	Feasibility studies, investment, operating costs, commodity risk and resource economics.

7.4 Petroleum engineering

#	Proposed subject	Course summary
1	Petroleum Engineering Fundamentals	Petroleum operations from exploration through drilling, reservoir development and production.
2	Petroleum Reservoir Engineering	Rock-fluid properties, reservoir pressure, flow, recovery, material balance and simulation.
3	Petroleum Production Engineering	Production systems, artificial lift, well performance and production optimization.
4	Well Completion and Stimulation Engineering	Casing, cementing, completions, stimulation operations and well integrity.
5	Formation Evaluation and Well Logging	Borehole logs, porosity, permeability, saturation and formation interpretation.
6	Petroleum Processing and Surface Facilities	Oil/gas/water separation, gathering, transport, storage and processing systems.
7	Petroleum Engineering Data Analytics	Production and reservoir data handling, predictive methods and digital field tools.
8	Energy Resources and Sustainable Development	Geothermal energy, carbon management, energy transitions, economics and sustainability.

7.5 Civil and geotechnical engineering

#	Proposed subject	Course summary
1	Statics and Mechanics of Materials	Equilibrium, internal force, stress, strain, deflection and structural materials.
2	Structural Analysis	Load effects in beams, frames, trusses and statically indeterminate structures.
3	Reinforced Concrete and Steel Structures	Design and performance of reinforced concrete and structural steel systems.
4	Construction Materials and Technology	Concrete, metals, aggregates and rock materials; testing, production and construction practice.
5	Foundation Engineering	Shallow/deep foundations, settlement, bearing resistance and stability.
6	Geotechnical Engineering	Soil and rock behavior, retaining structures, earthworks and slope stability.
7	Underground and Tunnel Engineering	Tunnel/underground space design, excavation methods and support systems.
8	Hydraulics and Water Resources Engineering	Open-channel flow, water systems, drainage, hydraulic works and resource management.
9	Transportation and Infrastructure Engineering	Roads, bridges, transport facilities, networks and infrastructure planning.
10	Construction Engineering and Project Management	Construction methods, cost, schedule, contracts, quality and safety management.
11	Building Information Modeling and Digital Construction	BIM, coordinated 3D models, construction information and digital project delivery.

7.6 Space science and geospatial technology

#	Proposed subject	Course summary
1	Fundamentals of Space Science	Solar system, space environment, celestial bodies and physical processes in space.
2	Planetary Science and Comparative Planetology	Origin, structure, geology and evolution of planets, moons and small bodies.
3	Atmospheric Physics and Space Weather	Atmosphere, ionosphere, magnetosphere, solar wind and effects on space operations.
4	Orbital Mechanics and Astrodynamics	Orbits, trajectories, orbital maneuvers and spacecraft motion.
5	Satellite Systems Engineering	Spacecraft architecture, power, communications, thermal systems and requirements.
6	Spacecraft Sensors and Instrumentation	Optical, radar and electromagnetic sensors; space instrumentation principles.
7	Space Mission Design and Systems Engineering	Mission objectives, requirements, spacecraft trades, system integration and risk.
8	Earth Observation and Remote Sensing	Satellite imagery, optical/radar imaging and Earth, resource and environmental monitoring.
9	Surveying, Geodesy and GNSS	Positioning, reference frames, measurements, satellite navigation and adjustment.
10	Photogrammetry, LiDAR and 3D Mapping	Image metrology, laser scanning, point clouds, terrain models and reconstruction.
11	Geographic Information Systems and Spatial Data Science	GIS, spatial databases, spatial analysis, cartography and geospatial modeling.
12	Space and Geospatial Data Analytics	Satellite/spatial data processing, statistics, machine learning and computation.

8. Recommended synthetic and interdisciplinary subjects

Synthetic subjects should integrate methods across departments and culminate in practical evidence: survey plans, interpreted maps, 3D models, engineering designs, technical reports or demonstrable space/geospatial systems. They should **replace** duplicated course content where practical, not simply enlarge the elective catalogue.

Integrated subject	Summary and expected scope
1. Integrated Mineral Exploration Engineering	Integrates geological mapping, economic geology, geophysics, geochemistry, drilling, satellite images and GIS to identify targets and prepare a resource exploration plan.
2. Integrated Subsurface Imaging and 3D Geological Modeling	Combines field geology, seismic/EM/gravity/magnetic data, boreholes, inverse methods and geostatistics into 3D/4D subsurface models.
3. Geological Engineering for Civil Infrastructure	Links engineering geology, structural geology, soils, rocks and civil design for tunnels, foundations, dams, bridges and site selection.
4. Satellite Remote Sensing for Mineral and Infrastructure Exploration	Uses multispectral/hyperspectral imagery, radar and GIS to support mineral targeting, construction planning and environmental monitoring.
5. Planetary Geology and Space Resource Exploration	Integrates geology, planetary science, geophysics, remote sensing and resource assessment to study planetary surfaces and potential extraterrestrial materials.
6. Integrated Drilling and Subsurface Engineering	Brings together drilling machinery, drilling fluids, rock mechanics, logging, petroleum operations and subsurface investigation.
7. Mineral Materials and Sustainable Construction Technology	Integrates mineralogy, processing, materials science and construction engineering to develop useful mineral-derived materials and products.
8. Geospatial Engineering and Smart Infrastructure	Combines GNSS, surveying, GIS, imagery, BIM and civil engineering for infrastructure digital twins and asset monitoring.
9. Geoheritage, Geotourism and Sustainable Regional Development	Combines geology, geomorphology, geospatial planning, tourism facilities, environmental assessment and regional development.
10. Environmental and Social Management of Resource Development	Links mining, hydrogeology, environmental assessment, project economics, community considerations, reclamation and safety.
11. Earth Observation and Space Mission Design	Combines orbital mechanics, satellite payloads, communications, Earth observation and geospatial analytics into a small mission concept.
12. Integrated Earth Resources and Infrastructure Design Project	A multi-major capstone covering target evaluation, drilling, mining or petroleum development, civil works, costs, environmental management and reporting.

8.1 Suggested combinations by academic area

Category	Subject #	Recommended participants
Resource exploration integration	1, 2, 4, 6	Geology, applied geophysics, applied geochemistry, drilling and petroleum geoscience.
Engineering integration	3, 6, 7, 10, 12	Civil, geological, drilling, mining, mineral processing and petroleum engineering.
Space/geospatial integration	4, 5, 8, 9, 11	Space science, surveying, GIS, remote sensing, Earth observation and geotourism.

For each degree, require one or two relevant integrated subjects, offer selected interdisciplinary electives and require one final-year multidisciplinary project. Students should not be expected to complete all twelve.

9. Recommended four-year educational sequence

Stage	Educational objective	Indicative activities
Year 1	Scientific foundations	Mathematics, chemistry, physics, computing, general education, technical communication and introductions to Earth/engineering/space.
Year 2	Disciplinary foundation	Geological/engineering/space core subjects, laboratory methods, field measurement and professional orientation.
Year 3	Professional specialization	Advanced major courses, field/laboratory applications, industry-linked methods, design and interdisciplinary electives.
Year 4	Professional integration	Internship, advanced electives, integrated design, independent thesis and professional practice.

9.1 Mandatory practical experiences

- Geological mapping and core logging for geology/mineral exploration students.
- Survey design, data acquisition and interpretation for geophysics/geochemistry students.
- Drilling, mining, mineral processing and petroleum laboratory or field training for resource-engineering degrees.
- Geotechnical investigations, structural design and infrastructure projects for civil engineering students.
- Surveying, GNSS, photogrammetry, satellite data processing and systems projects for space/geospatial students.

9.2 Illustrative integrated senior project

Integrated Mineral Resource Exploration and Development Project: multidisciplinary student teams evaluate a selected site or dataset and produce one coordinated report.

Specialization	Project responsibility
Geology	Geological mapping, target generation and geological model.
Geophysics	Field survey plan, processing and interpretation.
Geochemistry	Sampling design, anomaly evaluation and quality control.
Drilling	Borehole design, operations plan and technical constraints.
Mining / mineral processing	Mine concept, process route, safety and production outlook.
Petroleum engineering	Well/reservoir planning where project context is hydrocarbon related.
Civil engineering	Access roads, foundations, site works, utilities and infrastructure.
Geospatial engineering	GIS base maps, remote sensing and 3D digital modeling.

The assessed deliverables should include a technical report, engineering drawings/models, indicative economic analysis, safety/environmental considerations and an oral defense.

10. Implementation plan and performance measures

10.1 Recommended institutional actions

Establish the faculty identity. Use three linked strategic directions: Earth Resources Exploration and Evaluation; Earth Resources Engineering and Infrastructure Development; Space Science, Earth Observation and Geospatial Technology.

Review staffing and resources. Match proposed divisions and degrees to academic expertise, student demand, teaching loads, laboratories, field sites, computing and accreditation requirements.

Reduce course overlap. Build a university-wide matrix of each module's owner, outcomes, prerequisites, majors served, practical contact hours and overlapping topics.

Embed digital and intelligent technologies. Use programming, statistics, 3D modeling, AI, image analysis, geospatial analytics, digital drilling, intelligent mining and digital infrastructure in appropriate majors.

Develop partnerships. Prioritize resource/geoscience and geoexploration partners, plus civil engineering and space/geospatial institutions; support joint field camps, staff exchange and capstone sponsorship.

10.2 Phased roadmap

Phase	Indicative period	Key work
Phase 1	0-6 months	Audit academic staffing, facilities, regulations, programs, duplicated courses and industry needs.
Phase 2	6-12 months	Approve departmental boundaries; assign course ownership; define laboratory and center governance.
Phase 3	Year 1-2	Design syllabi, degrees, outcomes, credit maps, prerequisites and integrated subjects; conduct external review.
Phase 4	Year 2-3	Introduce revised curriculum in stages with fieldwork, labs, industry internships and capstones.
Phase 5	Year 3-5	Evaluate program outcomes, partnerships, professional recognition and continuous improvement.

10.3 Suggested indicators

Area	Examples of measurable indicators
Curriculum quality	Percentage of courses with mapped learning outcomes; duplicated-credit reduction; student workload by semester.
Practical learning	Field/laboratory/design hours per graduate; capstone completion and external assessment.
Professional outcomes	Graduate employment by discipline, employer feedback, relevant certification and advanced study.
Research integration	Shared laboratory use, multi-department projects, externally supported student research.
Implementation feasibility	Qualified instructors, equipment readiness, program costs and compliance status.

11. Overall conclusions and decisions

The existing integrated faculty concept is strategically useful. The immediate priority is not to establish more narrowly named laboratories or degrees, but to clarify discipline ownership, give each major an appropriate professional core, and make interdisciplinary education a defining strength.

#	Decision recommendation
1	Keep Earth science and engineering together at faculty level, with space/geospatial technology as a third mission.
2	Establish five academic departments with clear ownership of teaching and professional outcomes.
3	Retain and repurpose six shared research laboratories; create a drilling and subsurface engineering center.
4	Use 12 detailed academic areas, with only eight or nine initially developed as independent degrees where feasible.
5	Remove excessive repetition, strengthen project-based learning and require an integrated fourth-year experience.

Proposed institutional identity

Faculty of Geoscience and Technology
Earth · Resources · Infrastructure · Space
A faculty that develops scientific understanding of the Earth and space and applies advanced technologies to resource discovery, responsible development, infrastructure and geospatial problem-solving.

This document is a strategic concept proposal, not a complete costed implementation plan or an accreditation submission. Formal adoption should follow staff and facilities audits, stakeholder consultation, legal review and detailed degree-by-degree curriculum design.

12. Official institutional references and scope

The following official university sources support the general benchmarking examples in this report. Proposed departments, curriculum subjects, credit allocations and implementation timelines are advisory recommendations rather than official programs of these universities.

[R1] MIT - Climate System Science and Engineering
https://climate-major.mit.edu/about/

[R2] University of Cambridge - Earth Sciences undergraduate
https://www.esc.cam.ac.uk/undergraduate

[R3] University of Oxford - Earth Sciences course information
https://www.earth.ox.ac.uk/undergraduate/course-information

[R4] ETH Zurich - Departments and academic organization
https://ethz.ch/en/the-eth-zurich/organisation/departments-and-competence-centres/departments.html

[R5] Peking University - School of Earth and Space Sciences, undergraduate requirements
https://sess2.pku.edu.cn/english/undergraduate/curriculumrequirement/index.htm

[R6] Jilin University - College of Geoexploration Science and Technology, education
https://gest.jlu.edu.cn/en/Education.htm

[R7] Colorado School of Mines - Schools and degree programs
https://sesp.mines.edu/programs/

[R8] Colorado School of Mines - Minerals and Subsurface Energy programs
https://smse.mines.edu/programs/

[R9] ETH Zurich - Earth and Planetary Sciences department
https://eaps.ethz.ch/en/department.html